METRIC

MIL-DTL-51027B
12 February 2010
SUPERSEDING
MIL-A-51027A
19 August 1968

DETAIL SPECIFICATION

ANTISETTING COMPOUND, DECONTAMINATING SLURRY, M2

Inactive for new design after 24 November 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of antisetting compound.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of this document are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil/.

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FEDERAL SPECIFICATION

O-S-635	-	Sodium Polyphosphates, Technical

(Copies of this document are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1193	-	Standard Specification for Reagent Water
ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

(Copies of these documents are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

USP-NF	-	U.S. Pharmacopeia-National Formulary

(Copies of this document are available online at http://www.usp.org/ or from the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852-1790)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Antisetting compound shall be manufactured from the following materials:

a. Citric acid, anhydrous, fine granular, conforming to the USP-NF.
b. Sodium tripolyphosphate, technical grade, conforming to type III of O-S-635.
c. Calcium oxide, technical grade, 90 percent pure minimum, 94 percent nonvolatile matter minimum, 95 percent minimum particle size retained on No. 7 sieve.

3.2 Chemical composition. Antisetting compound shall be a thoroughly blended mixture conforming to the characteristics of table I when tested as specified herein.

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TABLE I. Chemical composition.

Material	Percent by weight	Test paragraph
Citric acid, anhydrous	46.5 ±1.0	4.3.1
Sodium tripolyphosphate	50.5 ±1.0	4.3.2
Calcium oxide	3.0 ±0.2	4.3.3

3.3 Particle size. A minimum of 95 percent by weight of the antisetting compound shall pass through a No. 4 sieve when tested as specified in 4.3.4.

3.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirement specified herein is classified as conformance inspection (see 4.2).

4.2 Conformance inspection.

4.2.1 Lotting. A lot shall consist of the antisetting compound produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3)

4.2.2 Sampling.

4.2.2.1 For antisetting compound test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 0.5 kilograms (kg) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test.

Number of containers in batch or lot	Number of sample containers
2 - 25	2
26 - 150	3
151 - 1,200	5
1,201 - 7,000	8
7,001 - 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For test. Approximately equal portions of all of the specimens except one taken in 4.2.2.1 shall be thoroughly mixed to form a composited specimen of no less than 0.5 kg. The composited specimen and the remaining individual specimen shall be tested separately and in duplicate as specified in 4.3, and the duplicates averaged. Failure of the average of any test by either the composited or the individual specimen to meet specified requirements shall be cause for rejection of the lot represented.

4.3 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.3.1 Citric acid.

4.3.1.1 Potassium permanganate solution.

a. Dissolve 48 grams (g) of potassium permanganate in one liter of water and heat the solution to boiling, keeping the solution slightly below the boiling point for one hour.

b. Filter the solution through pure asbestos (free from organic matter) or through a sintered glass filter crucible. Protect the solution from evaporation, dust, and reducing vapors. Keep the solution in the dark or in diffused light. If in time manganese dioxide settles out, refilter the solution.

4.3.1.2 Apparatus. (See figure 1.)

a. The decomposition vessel A is a 500 milliliter (mL) Ehrlenmeyer flask fitted with a two-hole rubber stopper through which pass a small condenser C and a 200 to 250 mL capacity dropping funnel B. The upper end of the dropping funnel is closed with a one-hole rubber stopper carrying a drying tube D filled with ascarite resting on a small plug of glass wool.

b. The stem of the dropping funnel reaches nearly to the bottom of the flask and is bent upwards at its lower end to prevent entrance of carbon dioxide. The upper end of the condenser is connected to the purification train by means of short sections of fresh gum rubber or tygon tubing. In this, and other glass-to-glass connections, the two ends of the glass tubing should be in contact.

c. The bottle E is partially filled with 1-to-1 sulfuric acid saturated with silver sulfate. The U-tube F is filled with glass beads and enough concentrated sulfuric acid to cover the bend in the bottom of the tube. The second U-tube G contains anhydrous magnesium perchlorate. The U-tubes H and I are the absorption tubes and are each filled two-thirds full of ascarite and one-third full of anhydrous magnesium perchlorate as shown in the insert on figure 1. The last tube J contains one-half anhydrous magnesium perchlorate and one-half ascarite.

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d. From the last U-tube a rubber hose leads to a flow regulating device K and to a vacuum source. All the U-tubes are hung by copper or brass wire from a rod and made as compact as possible.

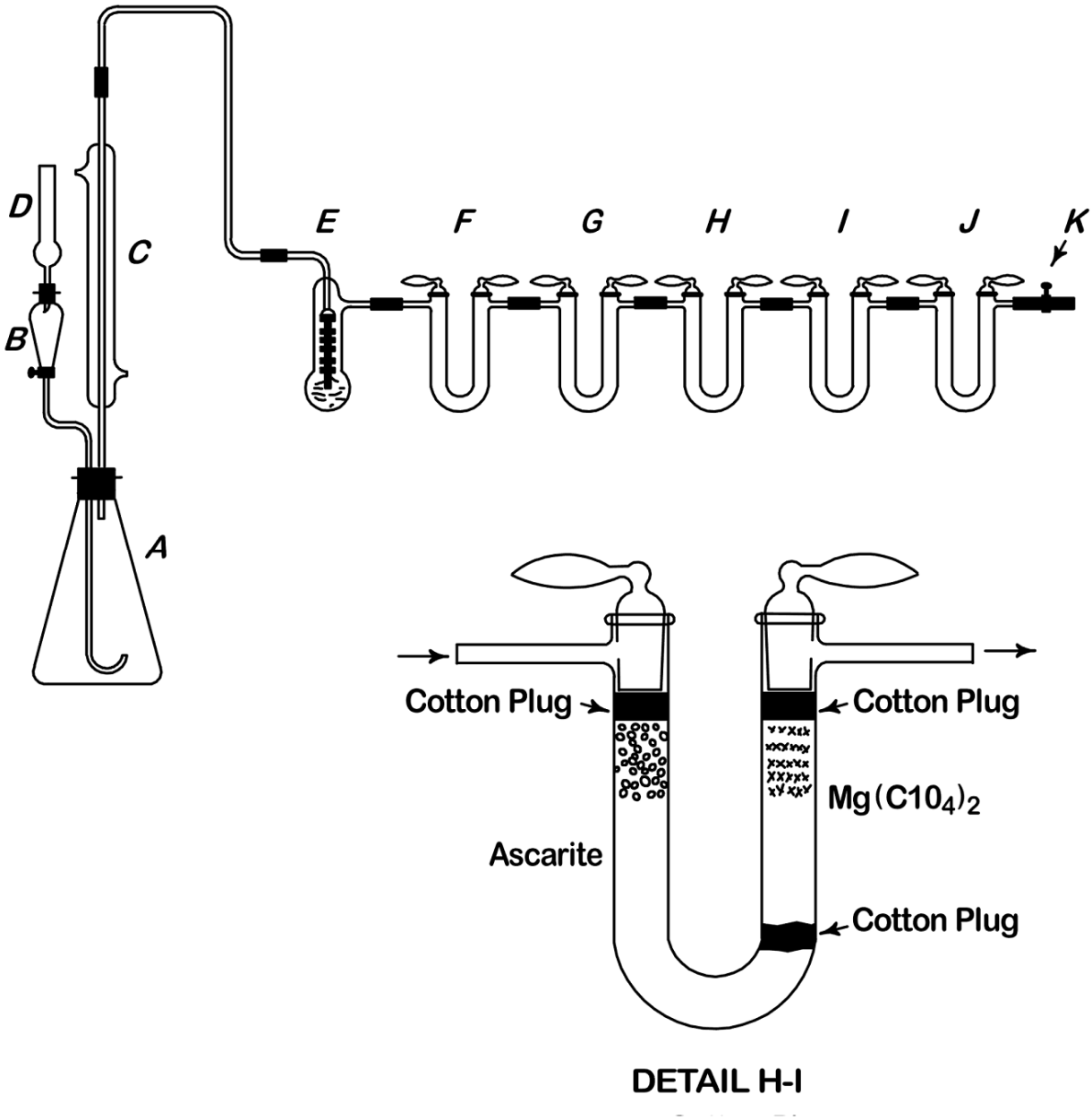


FIGURE 1. Apparatus for direct determination of carbon dioxide.

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4.3.1.3 Procedure.

a. First check the entire apparatus for leaks and to make certain all connections are gas tight. When the train has been proved gas-tight, draw a current of air through it of about two or three bubbles per second through E for 15 minutes. Then close the stopcocks on U-tubes J, I, H, and G in the order given.

b. Disconnect absorption tubes H and I and weigh them separately against a similar U-tube as a tare. Allow the tubes to stand in or near the balance case for one-half hour before weighing. Open the stopcock of each absorption tube and of the tare for an instant before weighing.

c. Replace the absorption tubes in the train, open stopcocks on tubes G, H, I, and J in the order given, and repeat the operation of drawing air through the tubes until the weight is constant to 0.5 milligrams (mg). When constant weight has been obtained, weigh 1 g of antisetting compound to the nearest 0.1 mg, dissolve in 75 mL of water, and add 6.0 mL of 1-to-1 sulfuric acid.

d. With all stopcocks in the absorption train closed, disconnect flask A and rinse the acidified antisetting compound solution into the flask. Replace flask A. Disconnect G from H. Open stopcocks on F and G. Apply suction to the open end of G for five minutes. Reconnect tubes G and H. Close stopcock on B. Remove D.

e. Add 200 mL of potassium permanganate solution, prepared as specified in 4.3.1.1, to B. Add enough water to displace any remaining air above the permanganate solution. Replace D.

f. Open stopcocks H, I, and J. Open stopcock B and start air flow through the absorption system at about three bubbles per second through E. Start water flowing through condenser C. Apply heat to A gently until the solution boils; then reduce heat. Keep the solution just below boiling for 15 minutes; then remove heat.

g. Continue air flow through the absorption train for at least 45 minutes after heat is removed. Close vacuum source K and stopcocks J, I, H, and G.

h. Remove and reweigh absorption tubes H and I separately against the respective tare, taking the same precautions as in the first weighings.

i. Calculate the percent citric acid as follows:

$$\text{Percent citric acid} = \frac{72.5 \cdot A}{W}$$

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Where:

A = Weight of carbon dioxide in grams.
W = Weight of specimen in grams.

4.3.2 Sodium tripolyphosphate.

4.3.2.1 Ammonium molybdate solution. Dissolve 200 g of ammonium molybdate in a mixture of 800 mL of water and 160 mL of 27 percent ammonium hydroxide.

4.3.2.2 Ammonium nitrate solution. Dissolve 55 g of ammonium nitrate in a mixture of 400 mL of nitric acid and 600 mL of water.

4.3.2.3 Wash solution. Dissolve 10 g of ammonium nitrate and 5 mL of nitric acid per liter of water.

4.3.2.4 Preparation of specimens. Crush 50 to 60 g of the specimen fine enough to pass a No. 20 sieve. Weigh to the nearest 0.01 g approximately 15 g of the crushed specimen, dissolve in water, and dilute to 1 liter. Transfer 100 mL of this solution to another liter flask and dilute to volume.

4.3.2.5 Procedure.

a. Transfer a 50 mL aliquot of the solution prepared as specified in 4.3.2.4 to a 600 mL beaker containing 100 mL of 1-to-1 nitric acid. Cover with a watch glass and boil gently for 30 minutes. Cool to room temperature.

b. To another 600 mL beaker, add 90 mL of ammonium nitrate solution prepared as specified in 4.3.2.2 and 45 mL of ammonium molybdate solution prepared as specified in 4.3.2.1. To this solution, maintained at 25 ±1 °C, add, with constant stirring, the treated aliquot of specimen solution. Allow the material to stand at 25 ±1 °C for 2 hours with occasional stirring.

c. Filter, using a tared Gooch crucible weighed to the nearest 0.1 mg and wash with the wash solution prepared as specified in 4.3.2.3. Dry the precipitate for 3 hours at 150 ±2 °C and weigh to the nearest 0.1 mg.

d. Calculate the percent sodium tripolyphosphate as follows:

$$\text{Percent sodium tripolyphosphate} = \frac{67.3 \cdot A}{W}$$

Where:

A = Weight of precipitate in grams.
W = Weight of specimen in the 50 mL aliquot in grams.

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4.3.3 Calcium oxide.

a. Weigh 6.0 g of the specimen to the nearest 0.1 mg into a 600 mL beaker. Add 100 mL of concentrated nitric acid and boiling chips (glass beads or Berl saddles).

b. Boil gently until oxidation ceases as evidenced by the evolution of white fumes. Add additional nitric acid if necessary. Do not allow the contents of the beaker to go to dryness.

c. Dilute the contents of the beaker to 100 mL and add several drops of methyl red indicator. Add concentrated ammonium hydroxide until within a few drops of the color change.

d. Dissolve 1.5 g of sodium oxalate in 200 mL of water, heating if necessary. Add the sodium oxalate solution to the beaker, stir well, and add concentrated ammonium hydroxide dropwise until the indicator changes to yellow. Bring the contents of the beaker to a boil to aid the formation of calcium oxalate.

e. Cool and filter through paper suitable for filtering fine precipitates. Wash the precipitate thoroughly with water.

f. Place a 500 mL iodine flask under the funnel and dissolve the calcium oxalate with hot 1.5 *N* sulfuric acid. Heat the flask contents to near boiling and titrate with 0.1 *N* potassium permanganate solution.

g. Calculate the percent calcium oxide as follows:

$$\text{Percent calcium oxide} = \frac{2.8 \cdot A \cdot B}{W}$$

Where:

A = Milliliters of potassium permanganate solution used.
B = Normality of potassium permanganate solution.
W = Weight of specimen in grams.

4.3.4 Granulation. Place an accurately weighed specimen of 100.0 g on a No. 4 sieve as identified in ASTM E11 (see 3.3) and assemble with a bottom pan. Cover and shake for three minutes by hand or by means of a mechanical shaker geared to produce 300 ±15 gyrations and 150 ±10 taps of a striker per minute. Weigh the portion passed by the sieve and calculate the percent passing.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Antisetting compound, decontaminating slurry, M2 is intended for use in retarding settling and setting in decontaminating slurry mixtures.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Packaging requirements (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method in ASTM E29, "Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications".

6.5 Subject term (key word) listing.

Calcium oxide
Citric acid
Decontamination
Mixture
Sodium tripolyphosphate

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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